

EXPERIMENT: Write a Prolog Program to suggest Dieting System based on Disease.

Aim

To write a Prolog program that suggests suitable food items based on a given disease using facts and logical rules.

Objectives

- To represent diseases and recommended foods using Prolog facts.
- To use rules to provide diet suggestions.
- To understand knowledge representation and logical inference in Prolog.
- To retrieve suitable diet recommendations using queries.

Algorithm

1. Start the program.
2. Define different diseases as facts.
3. Define recommended foods for each disease.
4. Define a suggest_diet rule.
5. Ask the user to enter a disease.
6. Search the knowledge base for the corresponding diet.
7. Display the recommended foods.
8. Stop the program.

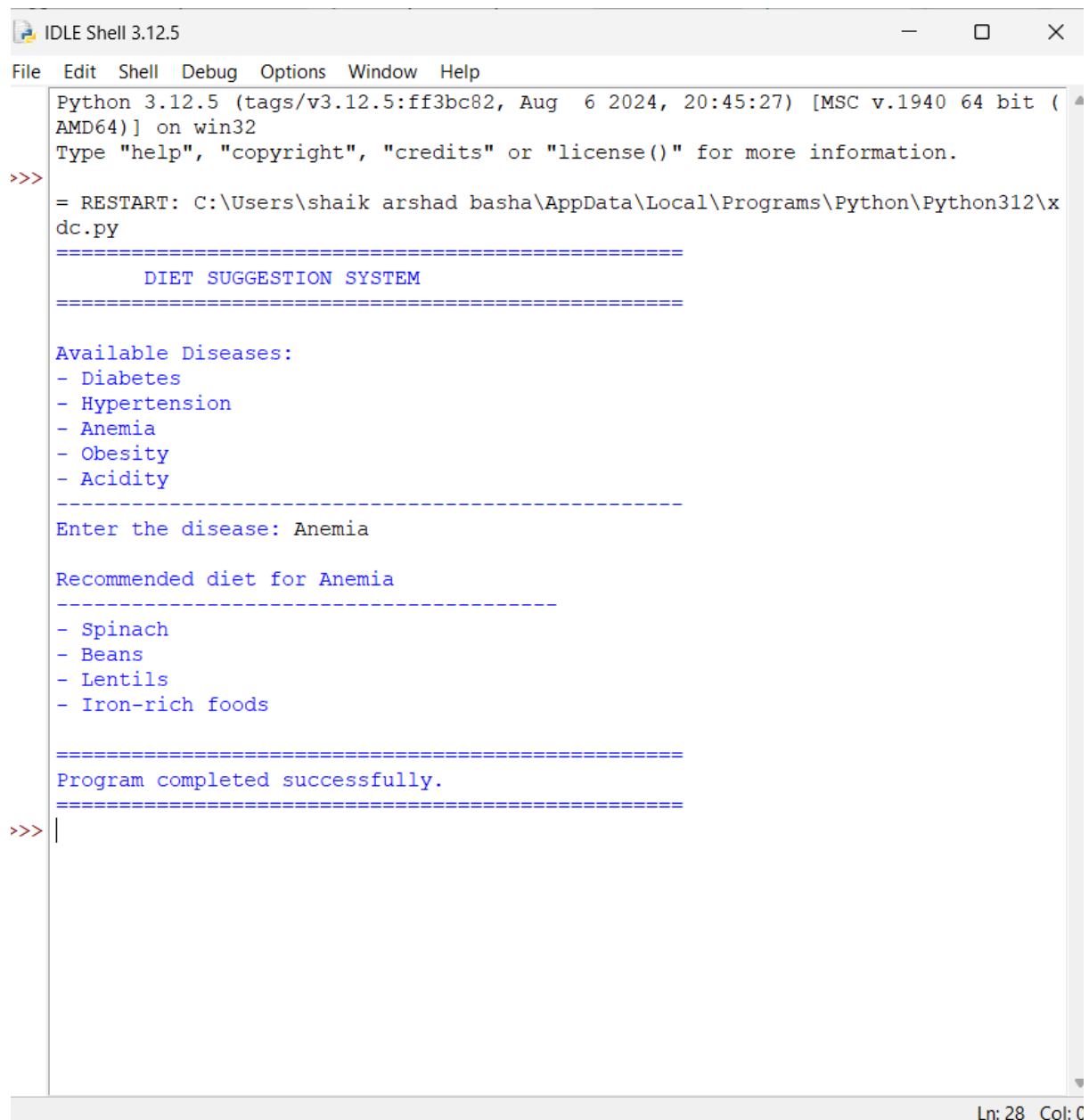
CODE

```
diet = {  
    "diabetes": [  
        "Vegetables",  
        "Whole grains",  
        "Beans",  
        "Nuts"  
    ],  
}
```

```
"hypertension": [  
    "Vegetables",  
    "Fruits",  
    "Whole grains",  
    "Low-salt food"  
],  
"anemia": [  
    "Spinach",  
    "Beans",  
    "Lentils",  
    "Iron-rich foods"  
],  
"obesity": [  
    "Vegetables",  
    "Fruits",  
    "Whole grains",  
    "Low-fat foods"  
],  
"acidity": [  
    "Banana",  
    "Oatmeal",  
    "Vegetables",  
    "Low-spicy foods"  
]  
}
```

```
def suggest_diet(disease):
    disease = disease.lower()
    if disease in diet:
        print("\nRecommended diet for", disease.title())
        print("-" * 40)
        for food in diet[disease]:
            print("-", food)
    else:
        print("\nDisease not found in the database.")
print("=" * 50)
print("    DIET SUGGESTION SYSTEM")
print("=" * 50)
print("\nAvailable Diseases:")
for disease in diet:
    print("-", disease.title())
print("-" * 50)
disease = input("Enter the disease: ")
suggest_diet(disease)
print("\n" + "=" * 50)
print("Program completed successfully.")
print("=" * 50)
```

OUTPUT



```
IDLE Shell 3.12.5
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shaik arshad basha\AppData\Local\Programs\Python\Python312\python.exe
dc.py
=====
DIET SUGGESTION SYSTEM
=====

Available Diseases:
- Diabetes
- Hypertension
- Anemia
- Obesity
- Acidity
-----
Enter the disease: Anemia

Recommended diet for Anemia
-----
- Spinach
- Beans
- Lentils
- Iron-rich foods

=====
Program completed successfully.
=====
>>> |
```

Ln: 28 Col: 0

Result

The Diet Suggestion System was successfully implemented in Prolog. The program provides food recommendations based on the disease entered by the user.

Conclusion

The program demonstrates how Prolog facts, rules, and queries can be used to create a simple knowledge-based diet recommendation system. The recommendations are educational examples and should not replace advice from a qualified healthcare professional.

